

## Question 1

Let  $f$  be strongly convex on  $S = \{x | f(x) \leq f(x^0)\}$ , i.e. there exists an  $m > 0$  such that

$$\nabla^2 f(x) \geq mI$$

for all  $x$  in  $S$ . Then

$$f(x) - p^* \leq (1/2m) \|\nabla f(x)\|_2^2$$

$f$  strongly convex  $\Rightarrow$  there exists  $m > 0$  s.t.  $\nabla^2 f(x) \geq mI, \forall x \in S$

Let  $x^*$  be the minimizer of  $f$ .  $\Rightarrow p^* = f(x^*), \nabla f(x^*) = 0$

$$\Rightarrow f(y) \geq f(x) + \nabla f(x)^T (y-x) + \frac{m}{2} \|y-x\|^2, \forall x, y \in S$$

$$\text{take } y = x^* \Rightarrow p^* = f(x^*) \geq f(x) + \nabla f(x)^T (x^* - x) + \frac{m}{2} \|x^* - x\|^2$$

$$\Rightarrow f(x) - p^* \leq \nabla f(x)^T (x - x^*) - \frac{m}{2} \|x - x^*\|^2 \leq \frac{1}{2m} \|\nabla f(x)\|^2$$

## Question 2

Let  $f$  be strongly convex on  $S = \{x | f(x) \leq f(x^0)\}$ . Suppose the exact line search is used in the gradient decent method, prove:

$$f(x^k) - p^* \leq c^k [f(x^0) - p^*]$$

let  $g_k = \nabla f(x_k)$ . use exact line search,  $x_{k+1} = x_k - t_k g_k$ .

where  $t_k$  minimizes  $f(x_k - t_k g_k)$

$$f \text{ strongly convex} \Rightarrow \text{from Q1, } f(x_k) - p^* \leq \frac{1}{2m} \|g_k\|^2, \|g_k\|^2 \leq 2m [f(x_k) - p^*]$$

Along the descent direction  $-g_k$ , exact line search gives

$$f(x_{k+1}) \leq f(x_k) - \alpha \|g_k\|^2$$

$$\Rightarrow f(x_{k+1}) - p^* \leq f(x_k) - p^* - \alpha \|g_k\|^2 \leq (1 - 2m\alpha) [f(x_k) - p^*]$$

$$\text{let } c = 1 - 2m\alpha, c \in (0, 1), \Rightarrow f(x_{k+1}) - p^* \leq c[f(x_k) - p^*]$$

$$\Rightarrow f(x_k) - p^* \leq c^k [f(x_0) - p^*]$$

### Question 3

Let  $f(x) = (1/2)(x_1^2 + \gamma x_2^2)$ , with  $\gamma > 0$ . Suppose that the gradient descent method is used with exact line search, starting at  $x^{(0)} = (\gamma, 1)$ . Prove

$$x_1^{(k)} = \gamma \left( \frac{\gamma - 1}{\gamma + 1} \right)^k, \quad x_2^{(k)} = \left( -\frac{\gamma - 1}{\gamma + 1} \right)^k$$

How about Newton descent?

$$\nabla f(x) = \begin{pmatrix} x_1 \\ \gamma x_2 \end{pmatrix}$$

$$x^{(k+1)} = x^{(k)} - t_k \nabla f(x^{(k)}) \Rightarrow x_1^{(k+1)} = (1 - t_k) x_1^{(k)}, \quad x_2^{(k+1)} = (1 - \gamma t_k) x_2^{(k)}$$

for quadratic function, the exact line search stepsize is

$$t_k = \frac{g_k^T g_k}{g_k^T Q g_k}, \quad Q = \begin{pmatrix} 1 & 0 \\ 0 & \gamma \end{pmatrix}, \quad g_k = \begin{pmatrix} x_1^{(k)} \\ \gamma x_2^{(k)} \end{pmatrix}$$

$$\Rightarrow t_k = \frac{(x_1^{(k)})^2 + \gamma^2 (x_2^{(k)})^2}{(x_1^{(k)})^2 + \gamma^3 (x_2^{(k)})^2}$$

$$\text{by induction, } x^{(0)} = (\gamma, 1). \quad x_1^{(k)} = \gamma \left( \frac{\gamma - 1}{\gamma + 1} \right)^k, \quad x_2^{(k)} = \left( -\frac{\gamma - 1}{\gamma + 1} \right)^k$$

$$\text{when } k=0, \quad x_1^{(0)} = \gamma, \quad x_2^{(0)} = 1.$$

$$\text{assume } x_1^{(k)} = \gamma r^k, \quad x_2^{(k)} = (-r)^k, \quad r = \frac{\gamma - 1}{\gamma + 1}$$

$$\Rightarrow (x_1^{(k)})^2 = \gamma^2 r^{2k}, \quad (x_2^{(k)})^2 = r^{2k}$$

$$\Rightarrow t_k = \frac{2}{\gamma+1} \Rightarrow x_1^{(k+1)} = (1-t_k)x_1^{(k)} = \gamma x_1^{(k)}$$

$$\Rightarrow x_1^{(k+1)} = \gamma r^{k+1}$$

$$x_2^{(k+1)} = (1-\gamma t_k)x_2^{(k)} = -\gamma x_2^{(k)} \Rightarrow x_2^{(k+1)} = (-\gamma)^{k+1}$$

$$\Rightarrow x_1^{(k+1)} = \gamma \left(\frac{\gamma-1}{\gamma+1}\right)^k, \quad x_2^{(k)} = \left(-\frac{\gamma-1}{\gamma+1}\right)^k$$

for Newton descent,

$$\nabla^2 f(x) = \begin{pmatrix} 1 & \\ & \gamma \end{pmatrix} \Rightarrow d_k = -(\nabla^2 f)^{-1} \nabla f = \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix}$$

$$\Rightarrow \text{with stepsize}=1, \quad x^{(1)} = (0, 0)$$